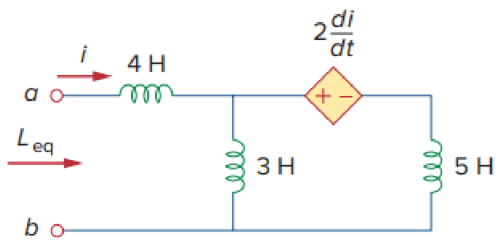
Problem

Determine *Leq* that may be used to represent the inductive network of Fig. 6.79 at the terminals.

Fig. 6.79:



Step-by-step solution

Step 1 of 4

Refer to Figure 6.79 in the textbook.

Represent the voltages and currents and redraw the circuit.

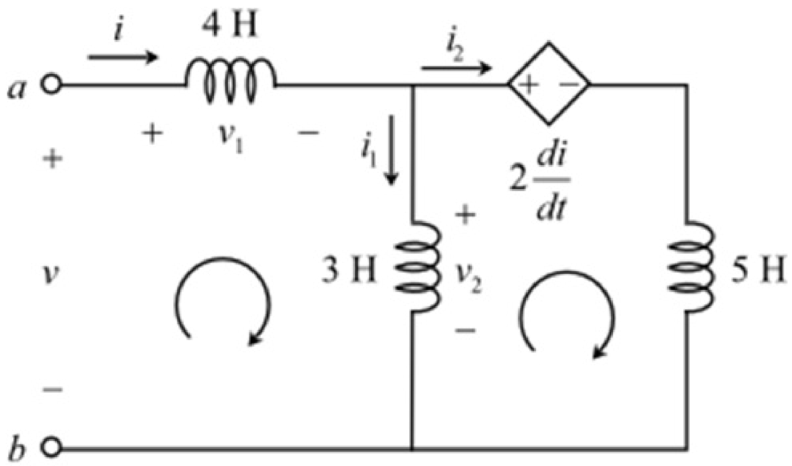


Figure 1

Step 2 of 4

Find the equivalent inductance of the circuit.

Apply Kirchhoff's voltage law to the first loop of the circuit.

$$\begin{aligned} v &= v_1 + v_2 \\ &= (4\text{ H})\frac{di}{dt} + v_2 \end{aligned}$$

Write the expression for the voltage v_2.

$$v_2 = (3\text{ H})\frac{di_1}{dt}$$

$$\frac{di_1}{dt} = \frac{v_2}{3}$$

Step 3 of 4

Apply Kirchhoff's voltage law to the second loop of the circuit.

$$-v_2 + 2\frac{di}{dt} + (5\text{ H})\frac{di_2}{dt} = 0$$

$$\begin{aligned}
v_2 &= 2\frac{di}{dt} + 5\frac{di_2}{dt} \\
&= 2\frac{di}{dt} + 5\frac{d}{dt}(i - i_1) \qquad (\text{since, } i = i_1 + i_2) \\
&= 2\frac{di}{dt} + 5\frac{di}{dt} - 5\frac{di_1}{dt} \\
&= 7\frac{di}{dt} - 5\frac{di_1}{dt}
\end{aligned}$$

Substitute the expression of the differentiation of i_1 in the equation.

$$v_2 = 7\frac{di}{dt} - 5\left(\frac{v_2}{3}\right) \qquad \left(\text{since, } \frac{di_1}{dt} = \frac{v_2}{3}\right)$$

$$v_2 + \frac{5}{3}v_2 = 7\frac{di}{dt}$$

$$v_2\left(1 + \frac{5}{3}\right) = 7\frac{di}{dt}$$

$$2.67v_2 = 7\frac{di}{dt}$$

Step 4 of 4

Simplify the expression further.

$$\begin{aligned}
v_2 &= \left(\frac{7}{2.67}\right)\frac{di}{dt} \\
&= 2.625\frac{di}{dt}
\end{aligned}$$

Substitute this expression of v_2 in the equation of v.

$$\begin{aligned}
v &= 4\frac{di}{dt} + v_2 \\
&= 4\frac{di}{dt} + 2.625\frac{di}{dt} \\
&= \frac{di}{dt}(4 + 2.625) \\
&= 6.625\frac{di}{dt}
\end{aligned}$$

The general expression for the input voltage is,

$$v = L_{eq}\frac{di}{dt}$$

Compare this expression with the voltage equation obtained. Therefore,

$$L_{eq} = 6.625\text{ H}$$

Therefore, the equivalent inductance of the circuit is, $\boxed{6.625\text{ H}}$.